

Practical 3

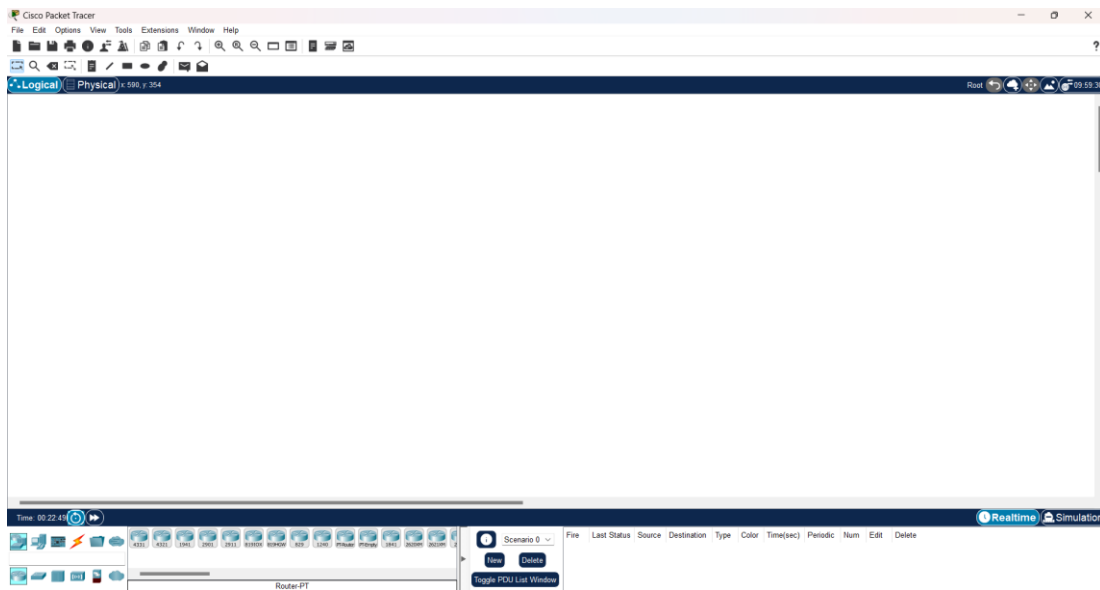
Simulate VLAN Configuration in Packet Tracer

Aim: To understand and configure VLANs for network segmentation and to observe communication between devices within and across VLANs.

Steps to Simulate VLAN Configuration:

1. Open Cisco Packet Tracer:

Launch the Cisco Packet Tracer application.



2. Add Network Devices:

Add a Switch:

From the *Network Devices* section, drag and drop a Switch (e.g., 2960 model) onto the workspace.

Add PCs:

From the *End Devices* section, drag and drop 4 PCs onto the workspace.

Add a Router:

Drag and drop a Router (e.g., 1941 model) from *Network Devices* to the workspace for inter-VLAN communication.

3. Cable the Network

Use a Copper Straight-Through Cable to connect:

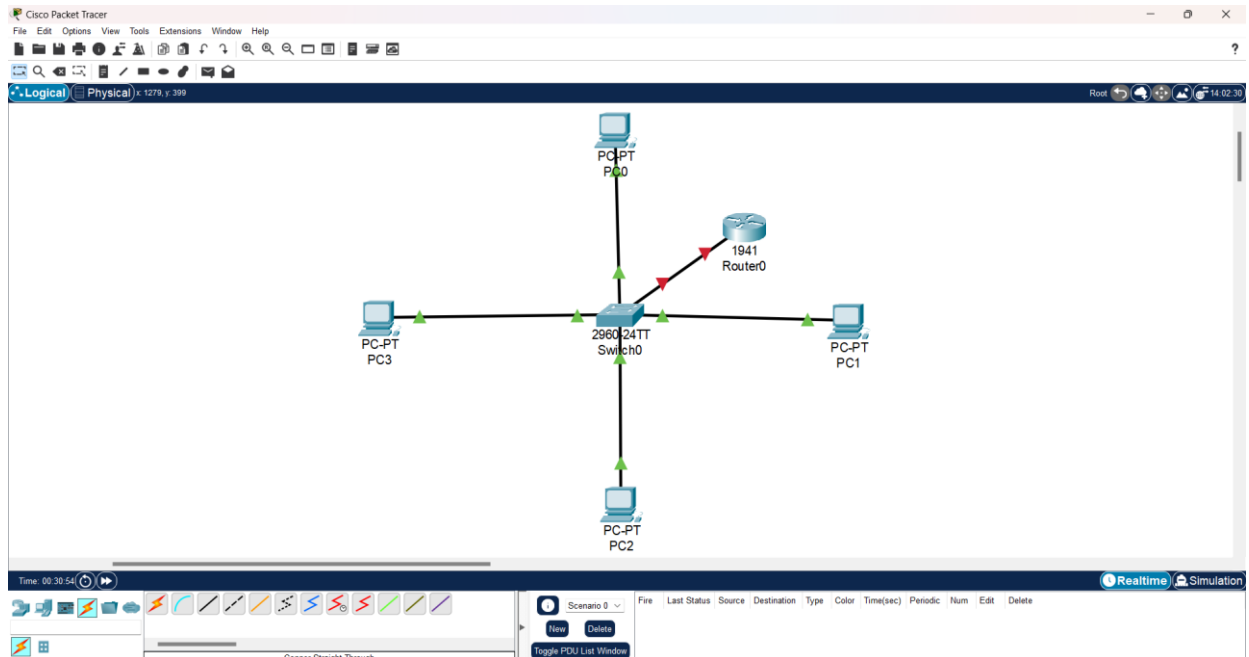
PC1 to FastEthernet0/1 of the switch.

PC2 to FastEthernet0/2 of the switch.

PC3 to FastEthernet0/3 of the switch.

PC4 to FastEthernet0/4 of the switch.

Switch to the Router's GigabitEthernet0/0 interface using a Copper Straight-Through Cable.



4. VLAN Creation on the Switch

Access the Switch CLI:

Click on the switch and go to the CLI tab.

```
Switch0
Physical Config CLI Attributes
IOS Command Line Interface
Motherboard serial number : FOC10093R12
Power supply serial number : AZS1007032H
Model revision number : B0
Motherboard revision number : B0
Model number : WS-C2960-24TT-L
System serial number : FOC1010X104
Top Assembly Part Number : 800-27221-02
Top Assembly Revision Number : A0
Version ID : V02
CLEI Code Number : COM3L00BRA
Hardware Board Revision Number : 0x01

Switch Ports Model          SW Version  SW Image
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* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
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Compiled Wed 26-Jun-13 02:49 by mnnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
```

Enter Configuration Mode:

CODE:

enable

configure terminal

Create VLANs:**Create VLAN 10 and VLAN 20:**

CODE:

```
vlan 10
name Sales
exit
vlan 20
name Marketing
exit
```

Assign VLANs to Ports:**Assign FastEthernet0/1 and FastEthernet0/2 to VLAN 10:**

CODE:

```
interface fastethernet0/1
switchport mode access
switchport access vlan 10
exit
interface fastethernet0/2
switchport mode access
switchport access vlan 10
exit
```

Assign FastEthernet0/3 and FastEthernet0/4 to VLAN 20:

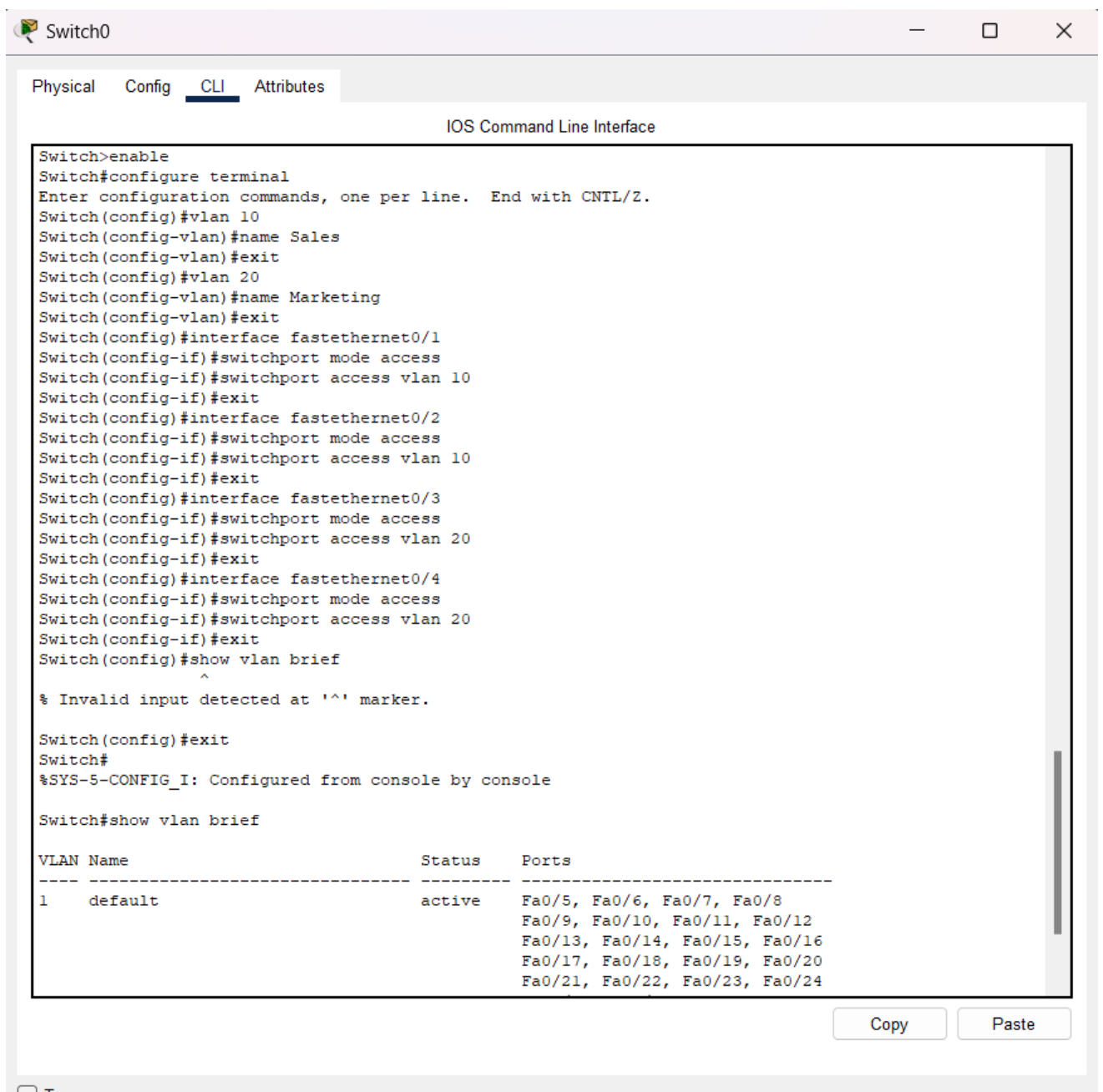
CODE:

```
interface fastethernet0/3
switchport mode access
switchport access vlan 20
exit
interface fastethernet0/4
switchport mode access
switchport access vlan 20
exit
```

Verify VLAN Configuration:

CODE:

```
show vlan brief
```



5. Configure the Router for Inter-VLAN Communication

Assign Subinterfaces:

Access the router CLI and configure GigabitEthernet0/0:

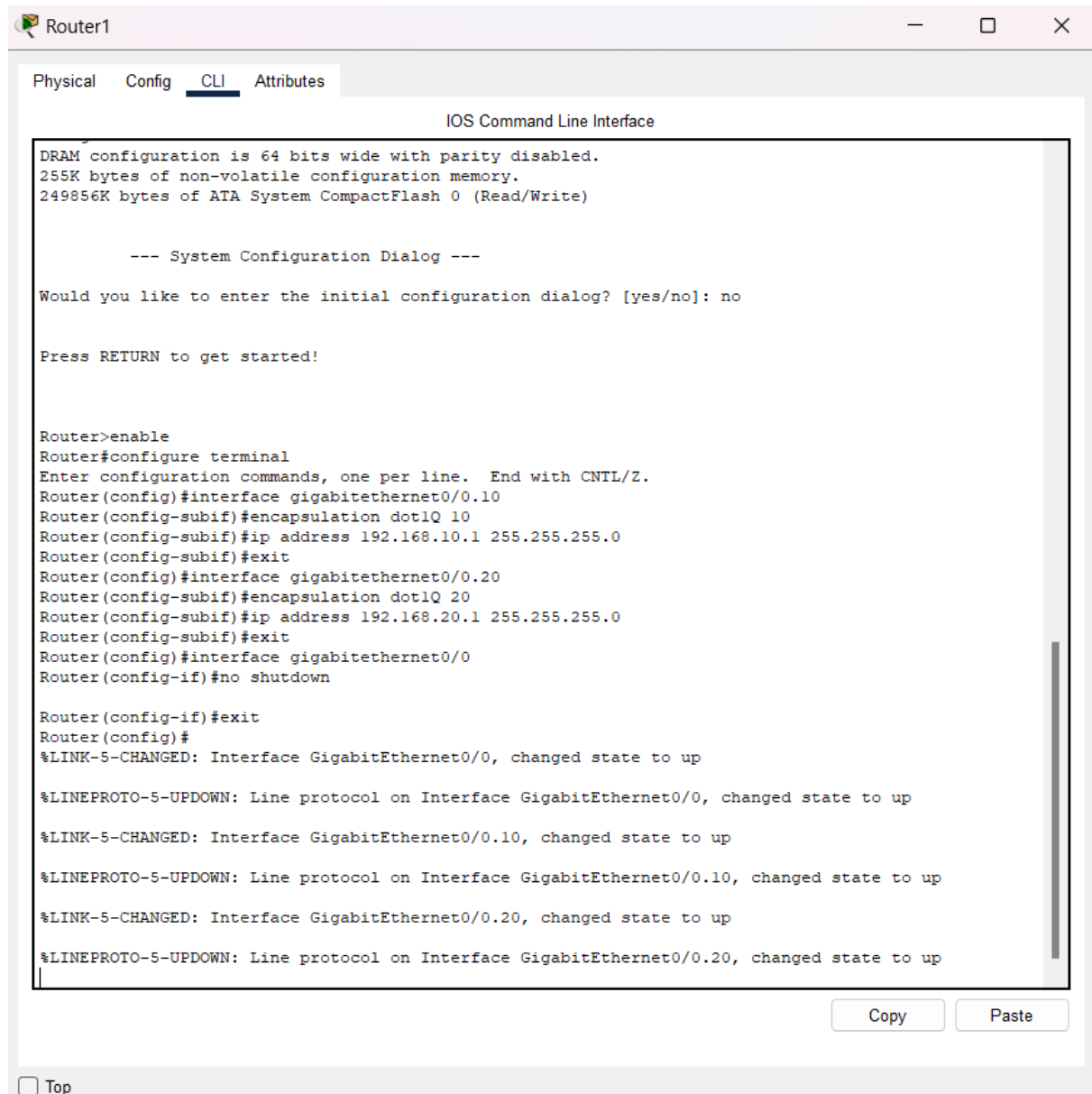
CODE:

```
enable
configure terminal
interface gigabitethernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
interface gigabitethernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

Activate the Router Interface:

CODE:

```
interface gigabitethernet0/0
no shutdown
exit
```



6. Configure IP Addresses on PCs

Assign static IP addresses to each PC as follows:

PC1: 192.168.10.2, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1

PC2: 192.168.10.3, Subnet Mask: 255.255.255.0, Gateway: 192.168.10.1

PC3: 192.168.20.2, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1

PC4: 192.168.20.3, Subnet Mask: 255.255.255.0, Gateway: 192.168.20.1

On each PC:

Open the Desktop tab.

Select IP Configuration.

Input the above IP details.

7. Test Communication

Ping PCs in the Same VLAN:

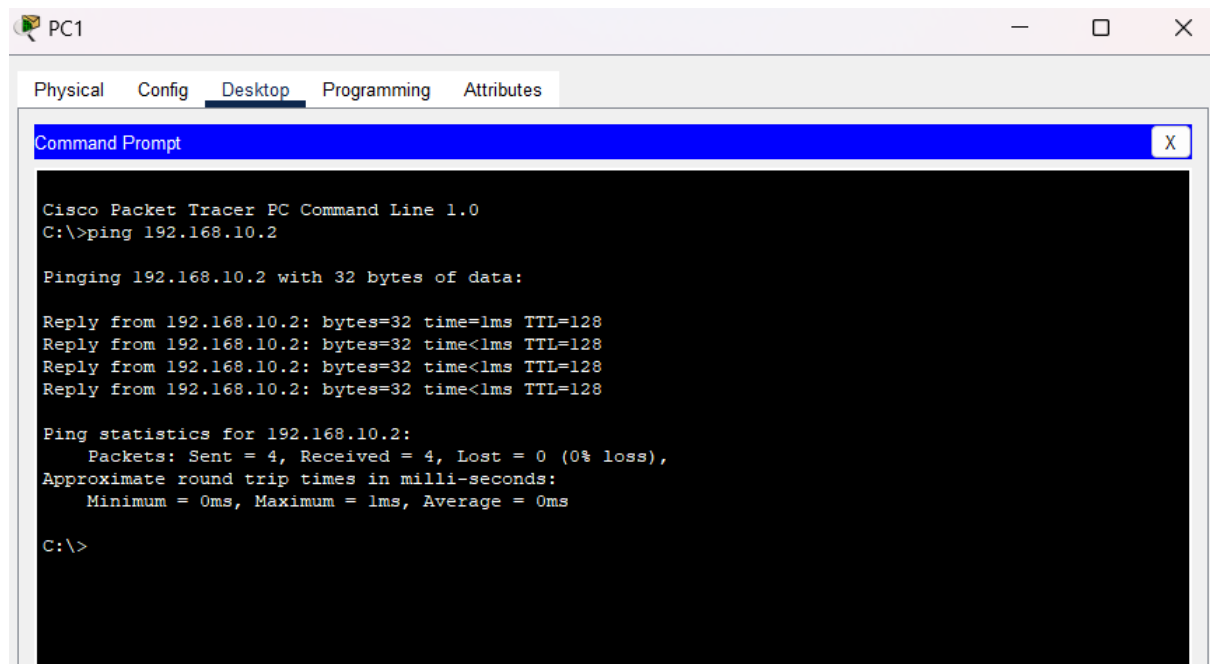
From PC1, ping PC2 (successful).

From PC3, ping PC4 (successful).

Ping PCs Across VLANs:

From PC1, ping PC3 (successful).

From PC2, ping PC4 (successful).



The screenshot shows a window titled 'PC1' with tabs for Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

8. Outcome:

VLANs were successfully created and configured on the switch.

Communication within the same VLAN was successful.

Inter-VLAN communication was facilitated using a router.

Students learned practical VLAN configuration and inter-VLAN routing.